

AVNITH VIJAYRAM

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EDUCATION

Carnegie Mellon University

May 2029

B.S. Computational Biology (School of Computer Science); additional major in Artificial Intelligence

GPA: 4.0/4.0

Thomas Jefferson High School for Science and Technology (TJHSST)

Jun 2025

SAT: 1580 · ACT: 36/36/36/36 · 13 college-level (AP/AV) courses in Math, Biology, and CS

GPA: 4.56

HONORS & AWARDS

- Putnam Mathematical Competition:** ranked 64th out of 4300+; **top scorer at CMU** (scored 60/120) Dec 2025
- Citadel Discover Program:** attended invitation-only 2-day program at Citadel for underclassmen Apr 2026
- ICPC Latin America Championship:** represented CMU; finished in 9th place Mar 2026
- Codeforces:** reached 2203 rating (Master, top ~1%) Sep 2025
- Regeneron Science Talent Search:** Named a Top 300 Scholar among 2,500 global candidates Jan 2025
- USA Computing Olympiad (USACO):** Achieved Platinum division, the highest tier Dec 2023

RESEARCH & PROJECTS

Research in Cardiovascular Medicine, Dr. Jennifer Kwan, Yale School of Medicine

Mar 2024 – Present

Identifying Biomarkers of Cardiotoxicity Risk for Cancer Patients

- Built a pipeline in Python and R to construct knowledge graph networks by applying gene coexpression inference on a cardiac transcriptomic dataset with 11,386 genes and 22 kinase inhibitors (KI) as cancer treatments
- Discovered novel causes behind the cardiotoxicity of KI treatments by modeling gene interactions using game theory and designing an algorithm that calculates a Nash equilibrium payoff score to predict gene expression strategies
- Identified biological pathways linked to myocarditis disease by applying Multi-Omics Factor Analysis (MOFA+) to scRNA-seq and CyTOF data to extract latent factors and genes; co-authored manuscript currently awaiting publication

GlioGrade: ML diagnostic tool for gliomas

May 2024 – Jun 2025

gliograde.org

- Achieved 93% accuracy at diagnosing gliomas from MRI images (700+ patients) with 3D Convolutional Neural Network models in PyTorch; used by neuropathologists at 3 clinics; advised by Dr. Kimmo Hatanpaa at UT Southwestern
- Built the full pipeline: preprocessing 3D scans (skull-stripping, artifact and rotation correction), mitigating class imbalance through sampling and weighting, and integrating the model into a local web app for clinicians

Data Intern (Volunteer), NOVA ScriptsCentral

Jan 2024 – Jun 2025

A non-profit for healthcare equity

- Cleaned and transformed 15 years of NSC's medication records using a Python/Pandas pipeline and used it to create a [data visualization dashboard](#) using Tableau to allow NSC to identify geographical areas for future expansion

EXPERIENCE

CMIMC Math Team, Head

Sep 2025 – Present

- Lead problemsetting for the CMIMC and CMWMC math competitions and manage the problem writing committee

USA Computing Olympiad, Coach

Oct 2025 – Present

- Proposed, selected, and prepared problems for monthly USACO contests with 5000+ participants

AlphaStar Academy, CS Instructor

Jun 2025 – Present

- Taught students and developed curriculum for USACO Silver-level algorithms and data structures concepts

TECHNICAL SKILLS

Languages: Python, C++, Java, R, SQL, LaTeX

ML / AI: PyTorch, TensorFlow, NumPy, Pandas, Convolutional Neural Networks (CNNs), Graph Neural Networks

Tools: Git, Linux, Jupyter, Tableau, Power BI

Domain: scRNA-seq, CyTOF, multi-omics (MOFA+), gene coexpression networks, algorithms, data structures, competitive programming